4MA1_1F_ch4_geometry_trigonometry

Nguyen Khac Kiem PhD

1. 4MA1/1FR_Summer_2024_Q23

The diagram shows a trapezium, $ABCD$

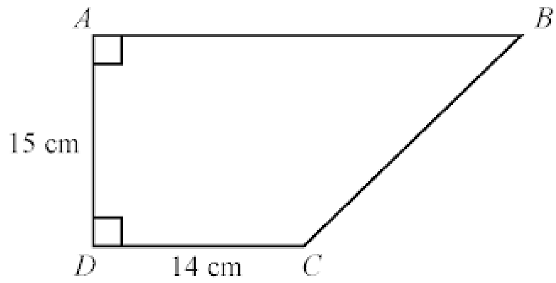


Diagram **NOT** accurately drawn

DAB and ADC are right angles.

$AD = 15$ cm $DC = 14$ cm

The area of the trapezium is 360 cm²

Work out the perimeter of the trapezium.

.... cm

(6)

2. 4MA1/1FR_Summer_2024_Q21

The diagram shows a solid wooden cylinder.

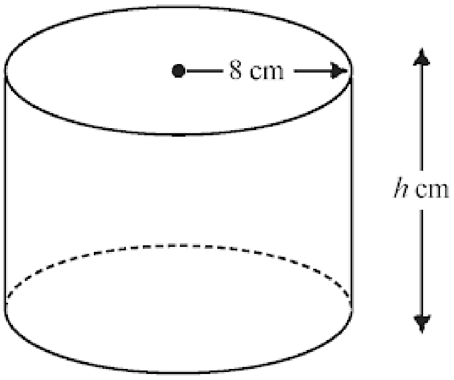


Diagram **NOT** accurately drawn

The cylinder has radius 8 cm and height h cm.
The volume of the cylinder is 1208 cm^3

(a) Work out the value of h
Give your answer correct to the nearest whole number.

h = ...
(2)

The density of the wood is 1.25 g/cm^3

(b) Work out the mass of the cylinder.
Give your answer in kilograms.

.... kilograms
(2)

3. 4MA1/1FR_Summer_2024_Q15

The diagram shows a box and a crate with a lid.

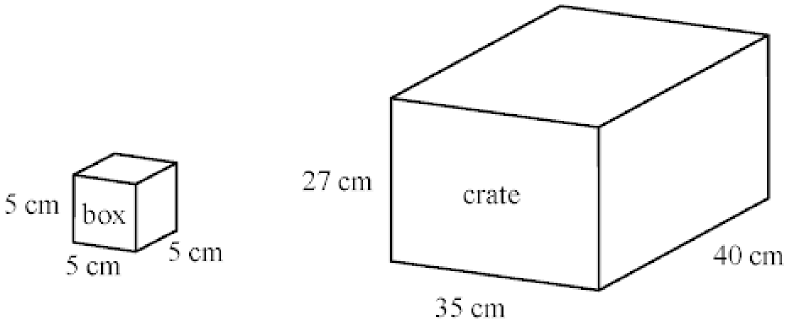


Diagram **NOT** accurately drawn

The box is a cube with sides of length 5 cm.
Tony has many boxes.

The crate is a cuboid with inside lengths of 27 cm, 35 cm and 40 cm.

Tony puts as many boxes as possible into the crate so that the lid will shut.

Work out the volume of space in the crate that is not filled up with boxes.

.... cm^3

(4)

4. 4MA1/1FR_Summer_2024_Q12

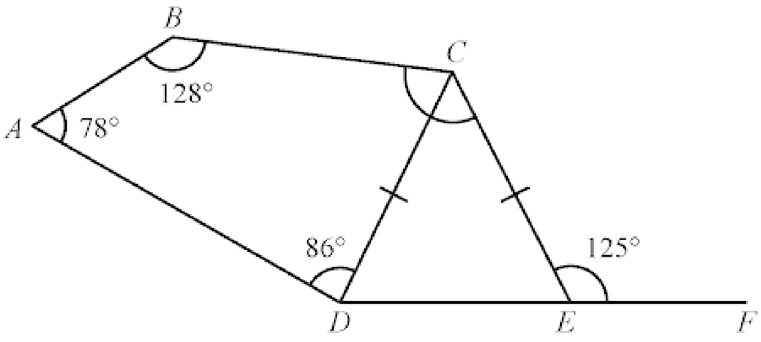


Diagram **NOT** accurately drawn

ABCD is a quadrilateral.
CDE is an isosceles triangle with *CD* = *CE*
DEF is a straight line.

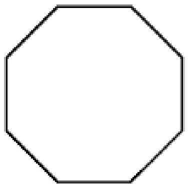
Work out the size of angle *BCE*
Give a reason for each stage of your working.

angle *BCE* = ... °

(5)

5. 4MA1/1FR_Summer_2024_Q6

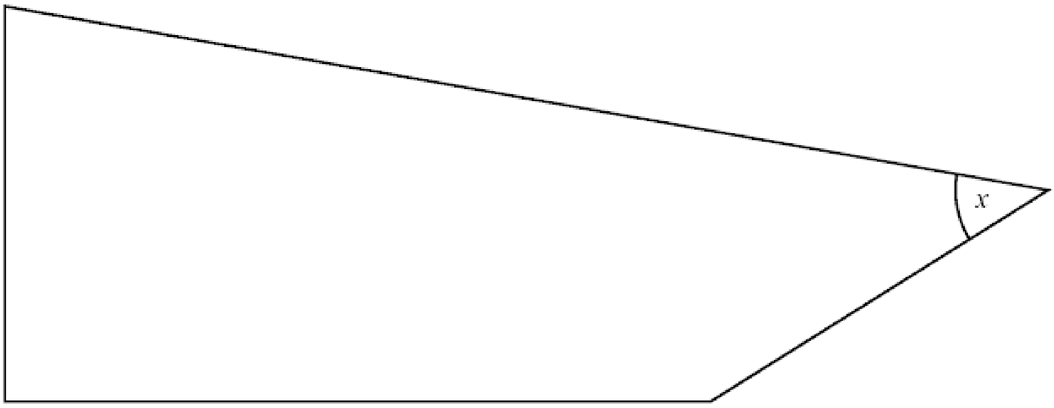
The diagram shows a regular octagon.



(a) Write down the number of lines of symmetry of a regular octagon.

(1)

Here is a shape.



(b) On the shape, mark with a letter R a right angle.

(1)

(c) On the shape, mark with a letter O an obtuse angle.

(1)

(d) Find, by measuring, the size of the angle marked x

..............................°

(1)

6. 4MA1/1F_Summer_2024_Q26

The diagram shows a roof support.

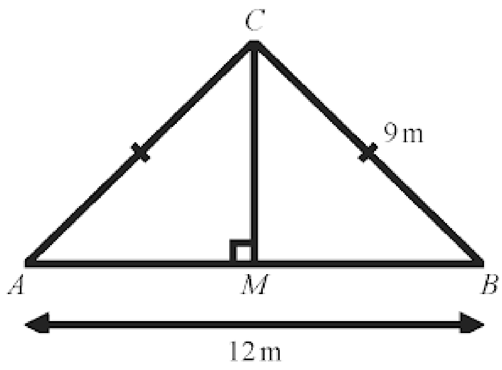


Diagram **NOT** accurately drawn

The roof support is made from four lengths of wood, AB, AC, BC and MC

$AC = BC = 9\text{ m}$ $\quad$ $AB = 12\text{ m}$

angle $AMC = 90°$

Lewis is going to buy lengths of wood to make the roof support.

The wood costs 21.50 euros per metre.
Each length of wood he buys has to be a whole number of metres.

Work out the total cost of the wood Lewis needs to buy.
Show your working clearly.

.... euros

(4)

7. 4MA1/1F_Summer_2024_Q20

In the diagram, $ABCDE$ is a regular pentagon.

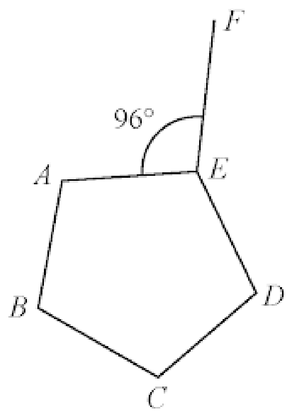


Diagram **NOT** accurately drawn

Angle $AEF = 96°$

Work out the size of the obtuse angle FED
Show your working clearly.

....°

(4)

8. 4MA1/1F_Summer_2024_Q15

The diagram shows a shape $ABCDE$ made from a right-angled triangle ABE and a square $BCDE$

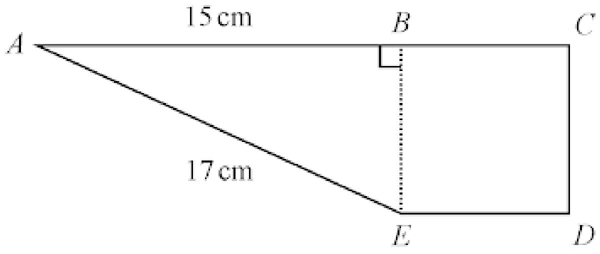


Diagram **NOT** accurately drawn

ABC is a straight line.

$AB = 15$ cm $\quad AE = 17$ cm

The perimeter of triangle ABE is 40 cm

Work out the area of the shape $ABCDE$

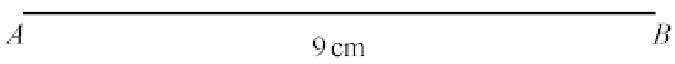
.... cm^2

(4)

9. 4MA1/1F_Summer_2024_Q9

ABC is an equilateral triangle with sides of length 9 cm.

Use a ruler and compasses only to **construct** the triangle ABC
The side AB has been drawn for you.
You must show all your construction lines.



(2)

10. 4MA1/1F_Summer_2024_Q6

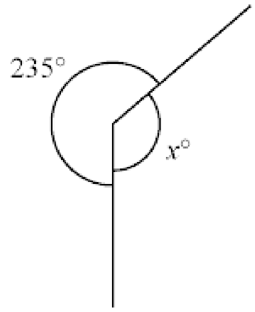


Diagram **NOT** accurately drawn

(a) (i) Work out the value of x

(1)

(ii) Give a reason for your answer.

(1)

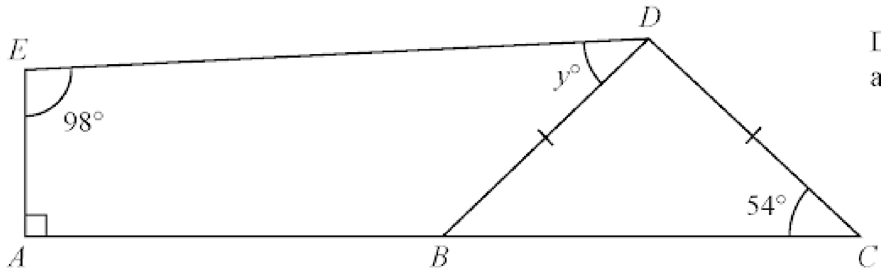


Diagram **NOT** accurately drawn

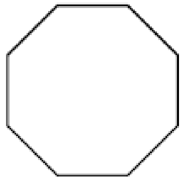
ABDE is a quadrilateral.
BCD is an isosceles triangle.
ABC is a straight line.

(b) Work out the value of y

(3)

11. 4MA1/1F_Summer_2024_Q4

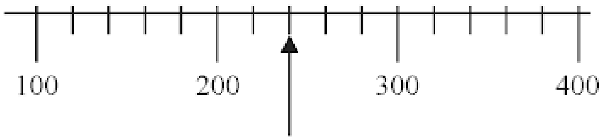
Here is a polygon.



(a) Write down the mathematical name of this polygon.

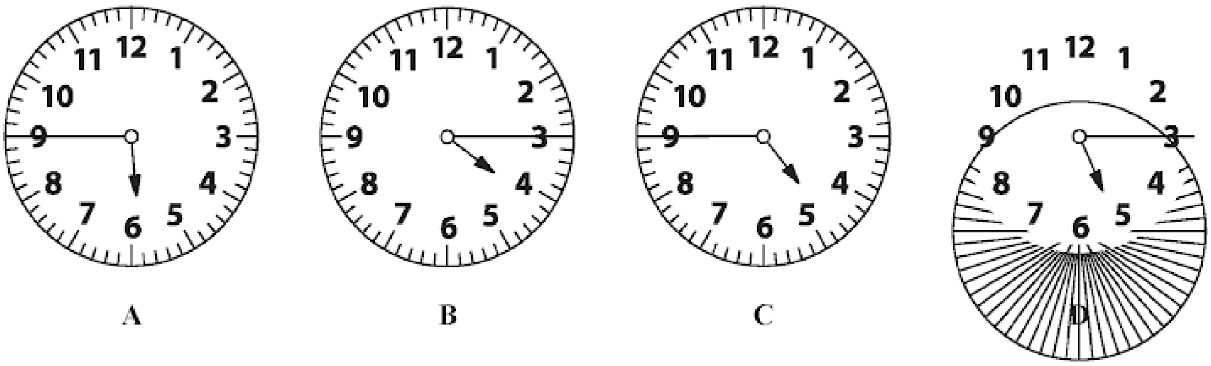
(1)

(b) Write down the number marked with the arrow.



(1)

Here are four clock faces.



(c) Write down the letter of the clock face that shows quarter to five.

(1)

(d) Complete the following sentence by writing a suitable metric unit on the dotted line.

The height of the Eiffel Tower is 300 ....

(1)

12. 4MA1/1FR_Winter_2023_Q25

R and T are points on a circle, centre O

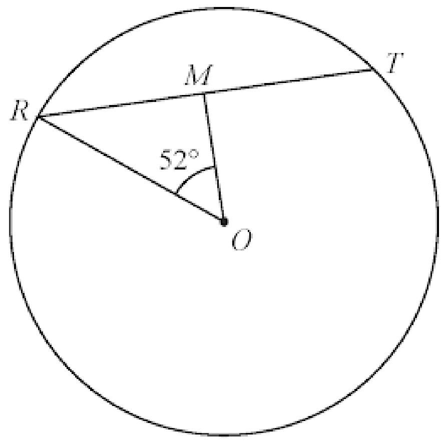


Diagram **NOT** accurately drawn

$RT = 12$ cm
M is the midpoint of RT
Angle $ROM = 52°$

Work out the area of the circle.
Give your answer correct to 3 significant figures.

.... cm^2

4 marks

13. 4MA1/1FR_Winter_2023_Q24

The diagram shows an isosceles triangle ABC

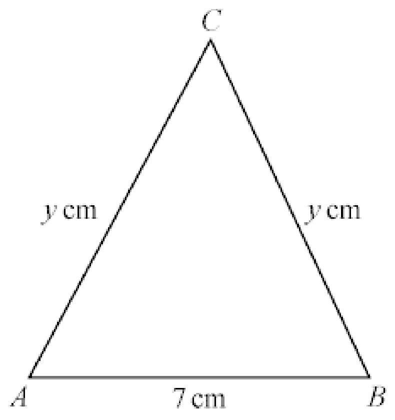


Diagram **NOT** accurately drawn

$AB = 7\text{ cm}$ $AC = BC = y\text{ cm}$

The area of the triangle is 42 cm^2

Work out the value of y

$y =$ ....

4 marks

14. 4MA1/1FR_Winter_2023_Q16

The diagram shows a solid triangular prism.

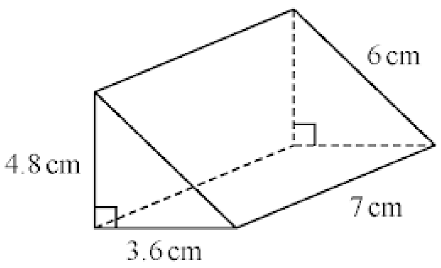


Diagram **NOT** accurately drawn

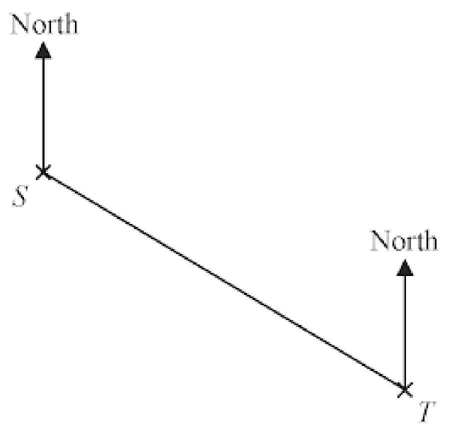
Work out the **total** surface area of the triangular prism.
Give your answer correct to 3 significant figures.

.... cm^2

3 marks

15. 4MA1/1FR_Winter_2023_Q14

The accurate scale drawing shows the positions of two lighthouses, S and T



The scale of the drawing is 1 cm to 2 km

(a) Find, by measuring, the bearing of lighthouse T from lighthouse S

....°

(1)

A boat is on a bearing of 084° from S
The boat is 13 km from T

(b) On the diagram, mark with a cross (×) the position of the boat.
Label the cross B

(3)

16. 4MA1/1FR_Winter_2023_Q13

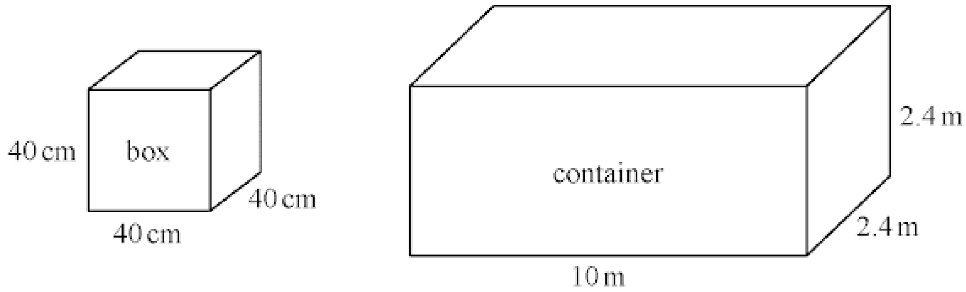


Diagram **NOT** accurately drawn

Tom puts boxes into a shipping container.

The container is a cuboid 10 metres by 2.4 metres by 2.4 metres.
Each box is a cube of side 40 centimetres.

Work out the greatest number of these boxes that Tom can put into the container.

....

3 marks

17. 4MA1/1FR_Winter_2023_Q9

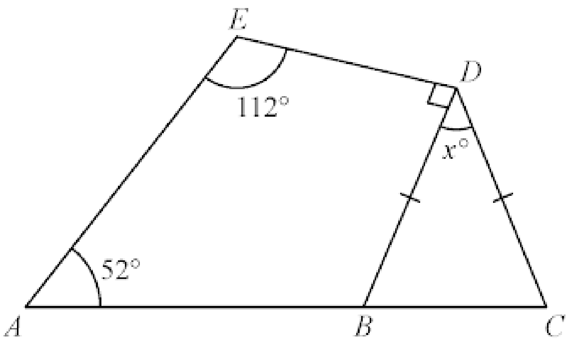


Diagram **NOT** accurately drawn

BCD is an isosceles triangle with $BD = CD$
ABC is a straight line.
$ABDE$ is a quadrilateral.

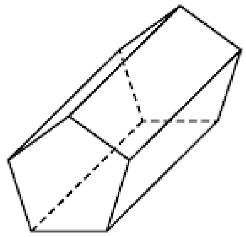
Work out the value of x
Give a reason for each stage of your working.

$x =$ ....

4 marks

18. 4MA1/1FR_Winter_2023_Q3

(a) Write down the mathematical name of this 3-D shape.



....
(1)

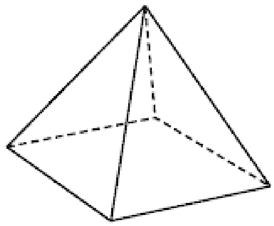
(b) (i) How many faces does this shape have?

....
(1)

(ii) How many vertices does this shape have?

....
(1)

Here is a different 3-D shape.



Marie makes a model of the shape.
She uses a length of wire to make each edge of the model.
Each edge of the model is 5 cm long.

Marie has 70 cm of wire.

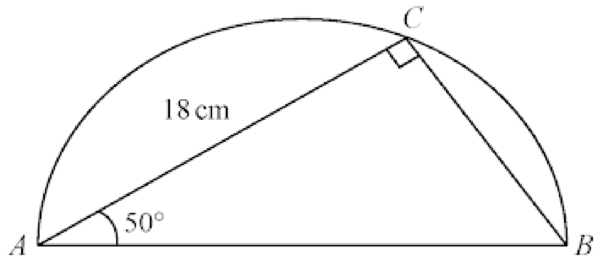
(c) What length of wire does she have left after making the model?

.... cm
(2)

19. 4MA1/1F_Winter_2023_Q23

The diagram shows a triangle ABC inside a semicircle.

Diagram **NOT** accurately drawn



A, B and C are points on the semicircle.

AB is the diameter of the semicircle.

Angle $ACB = 90°$
Angle $BAC = 50°$
$AC = 18$ cm

Work out the perimeter of the semicircle.
Give your answer correct to 2 significant figures.

20. 4MA1/1F_Winter_2023_Q19

The diagram shows a pentagon.

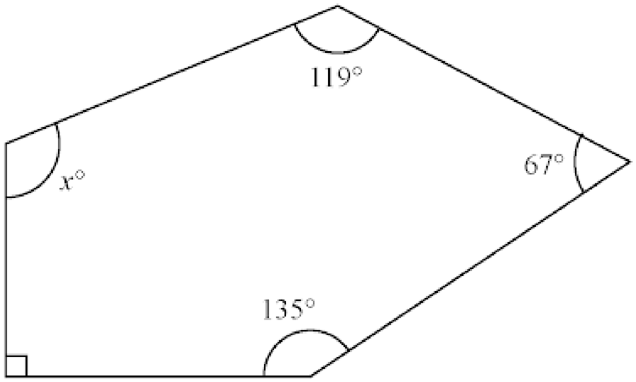


Diagram **NOT** accurately drawn

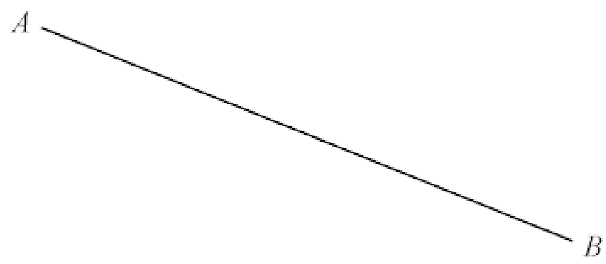
Work out the value of x

x = ...

3 marks

21. 4MA1/1F_Winter_2023_Q18

Use ruler and compasses only to construct the perpendicular bisector of line AB
You must show all your construction lines.



2 marks

22. 4MA1/1F_Winter_2023_Q15

Here is a floor plan of a stage.
The plan is formed from a triangle and a rectangle.

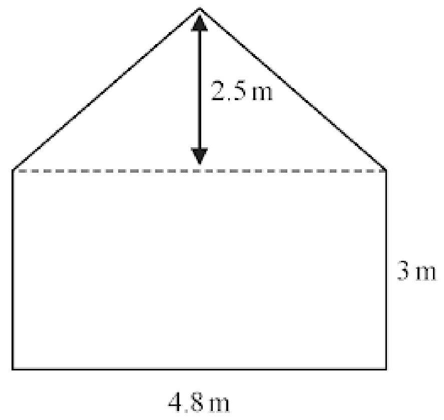


Diagram **NOT** accurately drawn

The stage manager is going to paint the floor.

One tin of paint covers an area of $1.8\,\text{m}^2$
One tin of paint costs \$16.40

Paint can only be bought in full tins.

The stage manager has \$190 to spend.

Does the stage manager have enough money to buy enough tins to paint all of the floor?
Show your working clearly.

5 marks

23. 4MA1/1FR_Summer_2023_Q27

The diagram shows a block of iron in the shape of a cuboid.

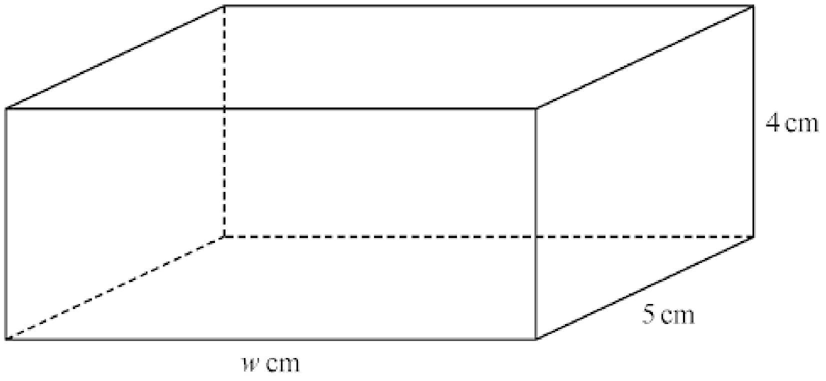


Diagram **NOT** accurately drawn

The block has length w cm, width 5 cm and height 4 cm

The density of iron is 7.8 g / cm^3
The mass of the block is 1950 g

Work out the value of w

w = ...

3 marks

24. 4MA1/1FR_Summer_2023_Q20

The diagram shows a shape made up of three semicircles, enclosing a right-angled triangle.

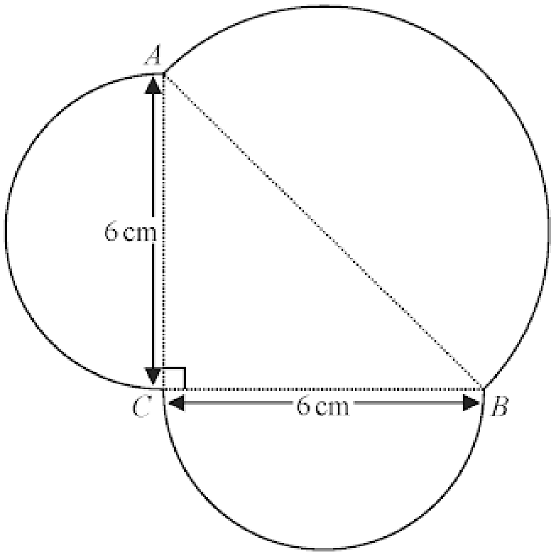


Diagram **NOT** accurately drawn

AB, BC and CA are each the diameter of a semicircle.

$BC = CA = 6$ cm.

Work out the perimeter of the shape.
Give your answer correct to one decimal place.

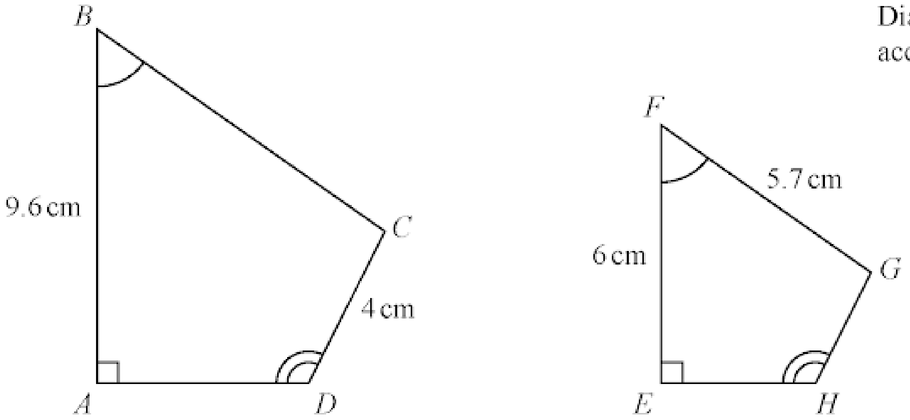
.... cm

5 marks

25. 4MA1/1FR_Summer_2023_Q19

$ABCD$ and $EFGH$ are similar quadrilaterals.

Diagram **NOT** accurately drawn



(a) Work out the length of GH

.... cm

(2)

(b) Work out the length of BC

.... cm

(2)

26. 4MA1/1FR_Summer_2023_Q13

27. 4MA1/1FR_Summer_2023_Q10

28. 4MA1/1FR_Summer_2023_Q3

29. 4MA1/1F_Summer_2023_Q20

30. 4MA1/1F_Summer_2023_Q13

31. 4MA1/1F_Summer_2023_Q7

32. 4MA1/1FR_Winter_2022_Q17

33. 4MA1/1FR_Winter_2022_Q8

34. 4MA1/1FR_Winter_2022_Q2

35. 4MA1/1F_Winter_2022_Q24

36. 4MA1/1F_Winter_2022_Q18

37. 4MA1/1F_Winter_2022_Q13

38. 4MA1/1F_Winter_2022_Q10

39. 4MA1/1F_Winter_2022_Q4

40. 4MA1/1FR_Summer_2022_Q27

41. 4MA1/1FR_Summer_2022_Q23

42. 4MA1/1FR_Summer_2022_Q12

43. 4MA1/1FR_Summer_2022_Q10

44. 4MA1/1FR_Summer_2022_Q5

45. 4MA1/1FR_Summer_2022_Q2

46. 4MA1/1F_Summer_2022_Q25

47. 4MA1/1F_Summer_2022_Q22

48. 4MA1/1F_Summer_2022_Q19

49. 4MA1/1F_Summer_2022_Q15

50. 4MA1/1F_Summer_2022_Q9

51. 4MA1/1F_Summer_2022_Q6

52. 4MA1/1FR_Winter_2021_Q25

53. 4MA1/1FR_Winter_2021_Q23

54. 4MA1/1FR_Winter_2021_Q20

55. 4MA1/1FR_Winter_2021_Q13

56. 4MA1/1FR_Winter_2021_Q10

57. 4MA1/1FR_Winter_2021_Q3

58. 4MA1/1F_Winter_2021_Q26

59. 4MA1/1F_Winter_2021_Q24

60. 4MA1/1F_Winter_2021_Q19

61. 4MA1/1F_Winter_2021_Q14

62. 4MA1/1F_Winter_2021_Q9

63. 4MA1/1F_Summer_2021_Q18

64. 4MA1/1F_Summer_2021_Q11

65. 4MA1/1F_Summer_2021_Q7

66. 4MA1/1FR_Winter_2020_Q23

67. 4MA1/1FR_Winter_2020_Q18

68. 4MA1/1FR_Winter_2020_Q10

69. 4MA1/1FR_Winter_2020_Q8

70. 4MA1/1FR_Winter_2020_Q3

71. 4MA1/1F_Winter_2020_Q22

72. 4MA1/1F_Winter_2020_Q14

73. 4MA1/1F_Winter_2020_Q13

74. 4MA1/1FR_Summer_2020_Q27

75. 4MA1/1FR_Summer_2020_Q18

76. 4MA1/1FR_Summer_2020_Q10

77. 4MA1/1FR_Summer_2020_Q9

78. 4MA1/1FR_Summer_2020_Q21

79. 4MA1/1F_Summer_2020_Q11

80. 4MA1/1F_Summer_2020_Q4

81. 4MA1/1F_Summer_2020_Q3

82. 4MA1/1FR_Winter_2019_Q25

83. 4MA1/1FR_Winter_2019_Q21

84. 4MA1/1FR_Winter_2019_Q18

85. 4MA1/1FR_Winter_2019_Q15

86. 4MA1/1FR_Winter_2019_Q3

87. 4MA1/1F_Winter_2019_Q20

88. 4MA1/1F_Winter_2019_Q19

89. 4MA1/1F_Winter_2019_Q17

90. 4MA1/1FR_Summer_2019_Q20

91. 4MA1/1FR_Summer_2019_Q16

92. 4MA1/1FR_Summer_2019_Q15

93. 4MA1/1FR_Summer_2019_Q16

94. 4MA1/1FR_Summer_2019_Q4

95. 4MA1/1F_Summer_2019_Q23

96. 4MA1/1F_Summer_2019_Q20

97. 4MA1/1F_Summer_2019_Q13

98. 4MA1/1F_Summer_2019_Q8

99. 4MA1/1F_Summer_2019_Q2

100. 4MA1/1FR_Summer_2018_Q21

101. 4MA1/1FR_Summer_2018_Q16

102. 4MA1/1FR_Summer_2018_Q12

103. 4MA1/1FR_Summer_2018_Q5

104. 4MA1/1FR_Summer_2018_Q4

105. 4MA1/1FR_Summer_2018_Q2

106. 4MA1/1F_Summer_2018_Q20

107. 4MA1/1F_Summer_2018_Q14

108. 4MA1/1F_Summer_2018_Q8

109. 4MA1/1F_Summer_2018_Q3